

National Physics Olympiad -2021
Category – C

Question 1: Little Prince's Planet

In this task you will estimate the (critical) number of atoms N_c (which contain both protons and neutrons in their nuclei) in an astronomical body for which the effects of gravity becomes significant for its existence, i.e. one cannot ignore its gravitational energy contribution anymore. We assume that all the atoms in it are all identical and carry atomic mass number A .

To facilitate our computation it will be convenient to use natural units where one sets the speed of light $c=1$ as well as the reduced Planck constant $\hbar=h/2\pi=1$. We will also need two other quantities (we write out c and $\hbar$ dependency explicitly for their expression) which are the electromagnetic fine-structure constant $\alpha = (e^2/\hbar c)$ and the gravitational “fine-structure” constant $\alpha_G = (G_N m_N^2/\hbar c)$.



Ignore the mass of the electron, m_e in the system when considering the gravitational effects. The average mass of a nucleon (proton or neutron) is taken to be m_N .

Let us now consider a spherical body of mass M which has a radius R . Both these quantities will scale as some powers of N (the number of atoms in the body) . In particular $R = N^p r_0$, where r_0 is a fixed length scale.

I. How does the surface area of this body scale with N ?

- a) $N^{1/2}$
- b) $N^{1/3}$
- c) $N^{1/4}$
- d) $N^{2/3}$
- e) $N^{-1/2}$

II. What would be an estimate for the gravitational potential energy, E_{Grav} for the system:

- a) $G m_N^2 A^2 N^{5/3} / r_0$
- b) $G m_N^2 A^3 N^{2/3} / r_0$
- c) $G m_N^2 A N^{5/3} / r_0^2$
- d) $G m_N^2 A^2 N^{2/3} / r_0$
- e) $G m_N^2 A^2 N^{2/3} / r_0^2$

III. The atomic binding energy for the whole system scales differently compared to the gravitational energy. If the atoms have a atomic number Z , what would be an estimate for the total atomic binding energy for the body (which is actually due to electromagnetism), E_{binding} ? (Hint: You can try to use your knowledge of the Bohr's model for hydrogen)

- a) $N^{2/3} Z \alpha^2 m_e^2$
- b) $N Z \alpha^2 m_e$
- c) $N Z^2 \alpha^2 m_e$
- d) $N Z^2 \alpha^3 m_e$
- e) $N Z \alpha m_e^2$

IV. For the gravitational contribution to dominate the atomic binding energy one needs $E_{\text{grav}} > E_{\text{binding}}$.

Solving this one gets the following the number of atoms must be greater than :

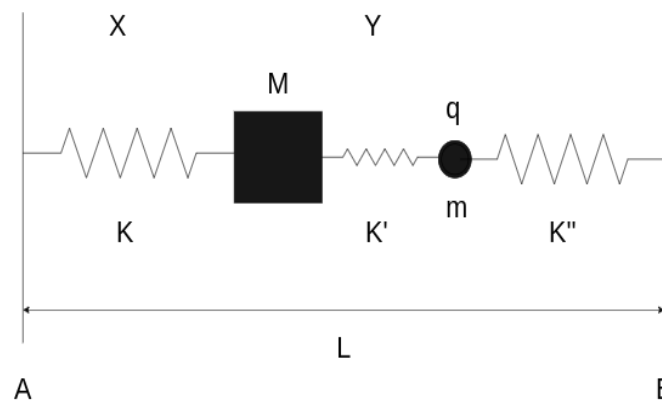
- a) $(Z/A^2)^{3/2} (\alpha/\alpha_G)^{3/2}$
- b) $(Z/A^2)^{2/3} (\alpha_G/\alpha)^{-1/2}$
- c) $(Z/A^2)^{2/3} (\alpha/\alpha_G)^{3/2}$
- d) $(Z/A^2)^{2/3} (\alpha/\alpha_G)^{-1/2}$
- e) $(Z/A^2)^{1/3} (\alpha/\alpha_G)^{1/2}$

V. Let the average value for this planet be $Z=10$ and $A=30$, what would be the order of magnitude for the numerical value of the critical mass

- a) 10^{25} kg
- b) 10^{23} kg
- c) 10^{24} kg
- d) 10^{20} kg
- e) 10^{27} kg

Question 2: (Rube Goldberg's Machine?)

Consider two blocks, one of a very large mass M and the other with mass m which are connected by springs as shown in the figure:



The second mass (m) also carries a charge q while the mass M is uncharged and made of an insulator. The masses are allowed to move only along the horizontal direction. The distance between the plates A and B as shown in the figure is fixed. Moreover there is a horizontal uniform electric field E that runs between A and B plates and points from left to right.

VI. What of the following expressions represent the potential energy of the system (ignoring the gravitational potential energy)?

- a) $\frac{1}{2} K X^2 + \frac{1}{2} K' Y^2 - q E Y$
- b) $\frac{1}{2} K X^2 + \frac{1}{2} K' Y^2 - q E (Y + X) + \frac{1}{2} K''(L - X - Y)^2$
- c) $\frac{1}{2} K X^2 + \frac{1}{2} K' Y^2 - q E Y + \frac{1}{2} K''(L - X - Y)^2$
- d) $\frac{1}{2} K X^2 + \frac{1}{2} K' Y^2 - q E (X + Y)$
- e) $\frac{1}{2} K X^2 + \frac{1}{2} K' Y^2 + \frac{1}{2} K''(L - X - Y)^2$

VII. What is the equilibrium values of X and Y (denoted as X_0 and Y_0 respectively) where X and Y are the displacements of the masses M and m respectively (see figure)?

- a) $(K'/(KK'+(K + K') K'')) (qE), (K'/(KK'+(K + K') K'')) (qE)$
- b) $qE/K, qE/K'$
- c) $(K'/(KK'+(K + K') K'')) (qE+ K'' L), (K/(KK'+(K + K') K''))$
- d) $K'/(K+K) L, K/(K+K') L$
- e) $(K'/(KK'+(K + K') K'')) (K'' L), (K'/(KK'+(K + K') K'')) (K'' L)$

VIII. Consider the case when $M=2m$ and $K=K'=K''$. What are the frequencies at which the shown masses M and m can oscillate freely?

- a) $\left[\frac{3}{2} \frac{K}{M} \right]^{1/2}, \left[\frac{3+\sqrt{3}}{2} \frac{K+qE/L}{M} \right]^{1/2}$
- b) $\sqrt{K/M}, 2\sqrt{K/M}$
- c) $\left[\frac{K-qE/L}{M} \right]^{1/2}, \left[\frac{K+qE/L}{M} \right]^{1/2}$
- d) $\left[3K/M \right]^{1/2}, \left[\frac{K-qE/L}{M} \right]^{1/2}$
- e) $\left[\frac{3+\sqrt{3}}{2} \frac{K}{M} \right]^{1/2}, \left[\frac{3-\sqrt{3}}{2} \frac{K}{M} \right]^{1/2}$

Answer Key to Category-C

- 1. (d)
- 2. (a)
- 3. (c)
- 4. (a)
- 5. (c)
- 6. (b)
- 7. (c)
- 8. (e)